

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION SP/2024)

CLASS: B.TECH
BRANCH: CSE/AIIML/ECE/EEE

SEMESTER : II
SESSION : SP/2024

SUBJECT: PH113 PHYSICS

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
 2. Attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates
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		CO	BL
Q.1(a)	What is Malu's law. Obtain the amplitude and intensity of a y-polarized light travelling in the x-direction passing through a polarizer with polarization axis at 30 degrees with respect to the y-axis.	[2]	1 2
Q.1(b)	Calculated the thickness of a doubly refracting crystal required to introduce a path difference of $\frac{\lambda}{4}$ between the e-ray and the o-ray when $\lambda = 6000 \text{ \AA}$, $\mu_o = 1.65$ and $\mu_e = 1.54$.	[3]	1 3
Q.2	Write down the conditions for maxima and minima in the interference pattern from parallel thin films. Now, consider white light reflected normally from a thin uniform glass slab of refractive index 1.65. Consecutive interference maximum and minimum are observed for $\lambda = 6000 \text{ \AA}$ and $\lambda = 4500 \text{ \AA}$, respectively (no other minima in between). Calculate the thickness of the glass slab.	[5]	1 3
Q.3(a)	Show that $\nabla \cdot (\nabla \times \vec{V}) = 0$ for an arbitrary vector field $\vec{V}(x, y, z)$.	[2]	2 1
Q.3(b)	Using Gauss's law, determine the electric field E due to infinite charge sheet with charge density σ .	[3]	2 2
Q.4	Obtain the boundary condition(s) for parallel/tangential and perpendicular/normal components of $\vec{D}$?	[5]	2 3
Q.5	Outline the fundamental differences between Galilean relativity and the special theory (Einsteinian) relativity.	[5]	3 4

:::21/02/2024 E:::